Campus Course & Records Manager (CCRM)

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1. Introduction

In today's academic environment, managing student records, courses, and enrollments efficiently is essential. Many institutions use large-scale software systems for this purpose, but smaller environments or learning contexts often lack simple tools to demonstrate how such systems work internally.

This project focuses on building a console-based Campus Course & Records Manager (CCRM) using Java. The goal is not to replace existing enterprise systems but to understand how such systems can be designed using core programming concepts.

The project demonstrates how basic operations such as adding students, managing courses, and tracking enrollments can be handled in a structured and modular way. It also emphasizes clean code organization and the use of object-oriented principles.

2. Problem Statement

In many learning scenarios, students study concepts like Object-Oriented Programming, file handling, and design patterns separately, without seeing how they come together in a real application.

The problem addressed in this project is:

How can we design a simple, structured system that manages student and course records while applying core programming concepts in a practical way?

The aim is to create a system that:

- is easy to understand
- demonstrates real-world structure
- applies multiple programming concepts together

3. Objective of the Project

The main objectives of this project are:

- To build a menu-driven application for managing students and courses
- To apply Object-Oriented Programming concepts such as classes, inheritance, and encapsulation
- To use design patterns like Singleton and Builder
- To implement file handling using Java NIO

- To organize code using proper package structure
- To provide a clear and interactive command-line interface

4. System Design and Architecture

The project follows a modular structure where different components are separated based on their roles.

Package Structure

- domain → contains core entities like Student, Course, Enrollment
- service → handles business logic
- io → manages file input/output and backup
- util → helper functions and validation
- config → application configuration using Singleton
- cli → main interface for user interaction

This separation ensures that the system is easy to maintain and extend.

5. Technologies and Concepts Used

This project uses Java SE and demonstrates several important concepts:

Object-Oriented Programming

Classes such as Student, Course, and Enrollment represent real-world entities. Encapsulation is used to protect data, and methods are used to define behavior.

Enums

Enums like Semester are used to define fixed values, improving readability and reducing errors.

Builder Pattern

Used for creating objects in a controlled and readable way.

Singleton Pattern

Used in configuration to ensure only one instance exists throughout the application.

File Handling (NIO)

Java NIO is used for efficient file operations, including reading and writing data.

Recursion

Utility functions demonstrate recursion for certain operations.

6. Implementation Details

The application starts with a menu-driven interface that allows the user to choose different operations.

When the program runs, the user is presented with options such as:

- List students
- List courses
- Enroll a student
- View enrollments
- Backup data

The system uses service classes to handle logic. For example:

- StudentService manages student-related operations
- CourseService manages courses
- EnrollmentService handles enrollments

The CLI acts as the bridge between the user and the system.

7. Screenshots and Output

7.1 Main Menu

```
aryan-mauryakant@LAPTOP-IJLI5FUP:/mnt/c/Users/Lenovo/OneDrive/Documents/ccrm_final_project$ bash compile_and_run.sh

--- CCRM Menu (by Aryan Mauryakant) ---
1) List students
2) List courses
3) Enroll student
4) List enrollments
5) Backup data
0) Exit
```

7.2 Listing Students

```
aryan-mauryakant@LAPTOP-IJLI5FUP:/mnt/c/Users/Lenovo/OneDrive/Documents/ccrm_final_project$ bash compile_and_run.sh

--- CCRM Menu (by Aryan Mauryakant) ---
1) List students
2) List courses
3) Enroll student
4) List enrollments
5) Backup data
0) Exit
1
Aryan Mauryakant - aryan.24BAI10142@vitbhopal.ac.in (24BAI10142)
```

7.3 Listing Courses

```
--- CCRM Menu (by Aryan Mauryakant) ---
1) List students
2) List courses
3) Enroll student
4) List enrollments
5) Backup data
0) Exit
2
CSx101: Programming in Java (4cr) - Shaiful Alom
```

7.4 Enrollment Display

```
--- CCRM Menu (by Aryan Mauryakant) ---
1) List students
2) List courses
3) Enroll student
4) List enrollments
5) Backup data
0) Exit
4
2 -> CSEx101 2026-03-30 N/A
```

7.5 Backup Operation

```
--- CCRM Menu (by Aryan Mauryakant) ---
1) List students
2) List courses
3) Enroll student
4) List enrollments
5) Backup data
0) Exit
5
Backup created: exports/backup_20260330_193550
Size: 0
```

8. Conclusion

The Campus Course & Records Manager project demonstrates how core programming concepts can be combined to build a functional system. Even though the application is simple, it reflects the structure of larger systems used in real environments.

The project successfully meets its objectives by providing a modular, menu-driven application that manages students, courses, and enrollments. It highlights the importance of clean design, proper organization, and practical implementation of theoretical concepts.

Overall, this project served as a valuable learning experience and helped bridge the gap between theory and real-world application.